# Health Data in South Korea

A practical introduction to survey, claims, and registry data

**Siho Park**
KDI School of Public Policy and Management
October 30, 2026

# Why Korean health data?

- **One insurer for the whole country.** National Health Insurance has covered essentially everyone since 1989, and the separate insurance societies were merged into a single insurer in 2000.
- **People do not disappear.** Records follow the same person across jobs, regions, and providers, without the gaps that arise when people switch insurers.
- **Screening is national and routine.** Most adults are invited to a free or nearly free health check every two years, which produces biomarkers for millions of people, not just for patients.

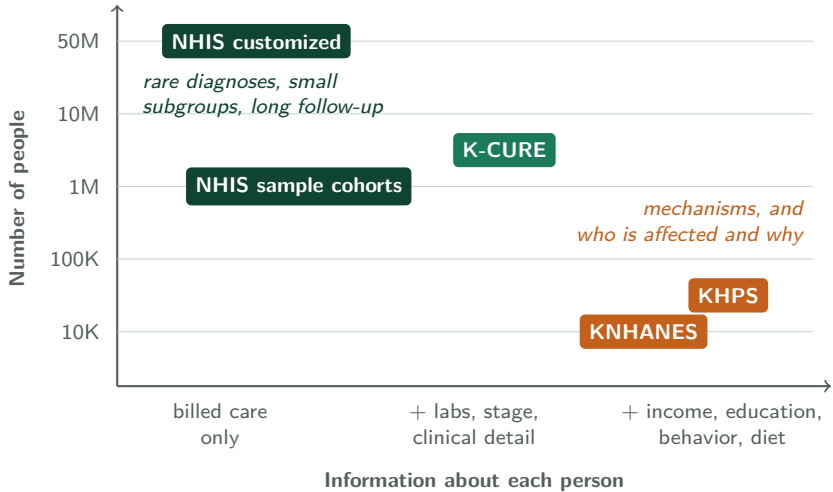
A rare combination: administrative data at population scale, alongside national surveys with the detail that administrative data lack.

## Four datasets in this talk

| | |
|---|---|
| **NHIS claims** | National health insurance claims data |
| **K-CURE** | Cancer registry data linked to claims, screening, and death |
| **KNHANES** | National survey with a physical examination and a nutrition interview |
| **KHPS** | Household panel survey with visit-level health care use and costs |

# The trade-off that organizes everything

Scale buys rare outcomes; surveys buy context



# Outline

1. NHIS claims data
2. K-CURE cancer data
3. KNHANES
4. Korea Health Panel Study
5. Choosing among them

**Two documentation sites**

I maintain English documentation for the two administrative sources, and this talk draws on it.

- NHIS Data Documentation
- K-CURE Data Documentation

SECTION 1

# NHIS claims data

## Where the data come from

Every record is a by-product of running the insurance program

- **Enrolling people** → eligibility records: insurance type, premium, disability, district of residence.
- **Paying claims** → claims records: diagnoses, procedures, drugs, and what the insurer and the patient each paid.
- **Running the screening program** → examination results: blood pressure, cholesterol, glucose, body measurements, and a lifestyle questionnaire.
- **Linking the death register** → date and cause of death from the national statistics agency.

**Why this framing matters**

The data measure what the program needed to record. Care that generates no claim, and income that sets no premium, simply are not there.

## What is inside

| Records | Unit | Main content |
|---|---|---|
| Eligibility (자격) | person-year | Insurance type, premium decile, disability, residence |
| Claims (진료) | claim and line | Diagnoses, procedures, drugs, spending, length of stay |
| Screening (건강검진) | examination | Blood pressure, lipids, glucose, BMI, smoking, drinking |
| Death (사망) | person | Date and cause of death |
| Long-term care (장기요양) | application, claim | Care grade assessment and use, from 2008 |
| Providers (요양기관) | provider-year | Type, size, and location of the facility |

Claims split into four linked tables: the claim itself, the service lines on it, the diagnoses on it, and outpatient prescriptions.

# The screening data are the underrated part

- Most adults are invited every two years; non-office workers every year.
- Gives **objective health measures for people who are not sick** — the counterfactual that claims data cannot provide.
- Includes a self-reported questionnaire on smoking, drinking, and exercise, so behavior and biomarkers sit on the same record.
- Repeated over two decades for the same people.

**Useful to know**

- Screening content changed in 2009 and 2018; the questionnaire items are not comparable across all years.
- Cancer screening is a separate program and is *not* in the standard claims or screening tables.

# Who is inside: the ready-made samples

| Database | Population | Size | Years |
|---|---|---|---|
| National Sample Cohort (표본코호트) | Enrolled throughout 2006, all ages | 1,000,000 | 2002–2023 |
| Health Screening Cohort (건강검진코호트) | Aged 40–79 and screened in 2002 or 2003 | 514,866 | 2002–2023 |
| Senior Cohort (노인코호트) | Aged 60–80 and enrolled in 2008 | 511,953 | 2002–2023 |
| Low-Risk Sample (저위험 표본) | Used medical care in 2014–2023 | 1,000,000 | 2014–2023 |

- The National Sample Cohort is the usual default: all ages, refreshed with newborns, and including Medical Aid beneficiaries.
- The Screening Cohort is richer on screening, but is a closed, aging cohort of people already screened — not the general population.
- **No household or family links** in any of them.

## Who is inside: the customized extract

- A **customized research DB** (맞춤형연구DB) is built from the full national database to match an approved protocol.
- It can cover **everyone who meets the study criteria**, not a sample — tens of millions of people.
- This is what makes rare outcomes feasible: a condition affecting one in ten thousand gives a handful of cases in a one-million sample and thousands in the full population.
- It is also the only route to diagnoses that are **masked in the sample databases**, including all mental and behavioral disorders, pregnancy and childbirth, and breast and genital cancers.
- The cost is a longer review, a higher fee, and analysis only at an NHIS analysis center.

# What you can study

## Natural fits

- Health care utilization and spending, at the visit, episode, or annual level
- Policy evaluation using age, date, or eligibility cutoffs built into the program
- Disease incidence, progression, and mortality over long horizons
- Provider behavior and regional variation in treatment
- Drug utilization and safety

## What the scale buys

- Rare diagnoses and rare adverse events
- Narrow subgroups: one age, one region, one occupation
- Precisely estimated effects at discontinuities, where only observations near the cutoff count
- Mortality as an outcome, which needs many person-years

## What it cannot tell you

- **No income and no education.** The premium decile stands in for income, but premiums come from wages for employees and from income and property for the self-employed, so the same decile means different things across groups.
- **Non-covered services are invisible** (비급여). Services patients pay for entirely out of pocket are never billed to the insurer, so any measure of total spending is incomplete, and incomplete by different amounts across conditions and years.
- **A diagnosis code is a billing code.** It records what justified a claim, not a confirmed diagnosis.
- **No families and no households**, in the sample databases.
- **Identifiers do not travel.** Each release has its own pseudonymized ID, so people cannot be followed across databases or linked to outside data.

# Getting access

## The process

- Apply at nhiss.nhis.or.kr
- IRB approval and protocol, **in Korean**
- Research summary read by a committee, **in Korean**
- Principal investigator must be a Korean national living in Korea
- Fee follows the period requested, not the days used; reduced 80% for a thesis, 50% for academic research

## Working with it

- You never receive the data; analysis runs on NHIS servers
- Sample databases: remote connection from your own computer
- Customized DB: in person at an analysis center
- **Both work only while you are physically in Korea**
- Results leave only after an export review

**For researchers based abroad**

Everything up to the analysis can be done online. The analysis itself needs a visit, so set the period of use around the trip.

SECTION 2

# K-CURE cancer data

## What K-CURE is

The national cancer registry, linked to everything else

- A data portal run by the Ministry of Health and Welfare that releases the **Korea Central Cancer Registry** (중앙암등록본부) joined to administrative records.
- The registry says **who was diagnosed, with which cancer, when, and at what stage**. The linked files say what care came before and after.
- The main product, the **Cancer Public Library** (암 공공 라이브러리), covers about 2.26 million registered cancer patients.

**Why not just use claims?**

In claims data a cancer is a billing code, with no stage, no histology, and an uncertain date. A registry record is a reported cancer with a diagnosis date and a stage — which is exactly what survival and early-detection research needs.

## What is inside

| Group | Content |
|---|---|
| Registry (RGST) | One row per registered cancer: site, histology, diagnosis date, SEER summary stage. Two cancers means two rows. |
| Death | Date and cause of death |
| Eligibility | Insurance type and premium, one row per person-year |
| Claims | Claim, service line, diagnosis, and outpatient prescription |
| Health checkup | Examination results and the lifestyle questionnaire |
| Cancer screening | Stomach, breast, colorectal, cervical, and liver, separately |
| COVID-19 | Confirmed cases and vaccinations |

All tables join on one person identifier. Cancer screening results are here, where they are *not* in the standard NHIS sample databases.

# Three ways to get it

| | **Customized** (맞춤형) | **Site sample** (표본) | **Staging sample** |
|---|---|---|---|
| Who | Every registered cancer, 2012–2022 | About 20% of one cancer site | About 10% of one site |
| Sites | All | Stomach, breast, colorectal, liver, lung, pancreas | Stomach, breast, colorectal, liver, lung |
| Stage | Summary stage | Summary stage | Tumor size, nodes, metastasis, grade, TNM |
| Variables | What you request | Fixed shorter list | Fixed list |
| Analyzed | At a safe-use center, in person | Remote virtual desktop | Remote virtual desktop |
| Wait | ∼3 months | ∼1 month | ∼1 month |

A study of thyroid, cervical, or hematologic cancer, or of several sites together, needs the customized extract.

# What you can study

## Questions it answers well

- Survival by stage, site, and year of diagnosis
- Whether screening shifts cancers toward earlier stages
- Treatment patterns and how they vary across providers and regions
- The cost of cancer care, and who bears it
- Socioeconomic and regional disparities in stage at diagnosis

## Limits to plan around

- Cancer patients only — there is no healthy comparison group
- Starts in 2012
- You cannot link further person-level data yourself; a linkage has to be built into the project from the start
- Some items, such as single-year age, are released only with a justification

Access conditions resemble the NHIS: Korean principal investigator, IRB approval, analysis in a controlled environment, and export review.

SECTION 3

# KNHANES

## Korea National Health and Nutrition Examination Survey

국민건강영양조사 — run by the Korea Disease Control and Prevention Agency

- A nationally representative survey, running annually, of roughly **10,000 people a year** across all ages.
- Three components collected for the same person:
  - **Health interview**: conditions, health care use, education, occupation, household
  - **Health examination**: blood pressure, height and weight, blood and urine tests, and depending on the year, vision, hearing, lung function, and bone density
  - **Nutrition survey**: a 24-hour dietary recall and food frequency, done at home
- **Cross-sectional**: a fresh sample each year, not a panel.

**The distinctive feature**

Measured values, not reported ones. You can see who has high blood pressure *and does not know it* — something no claims database can show.

# What you can study, and what you cannot

## Strengths

- National prevalence and long-run trends in obesity, diabetes, hypertension, smoking
- Diet and nutrition as an exposure, in detail available nowhere else
- Undiagnosed and untreated disease
- Socioeconomic gradients, since income and education are measured directly
- **Free and immediate**: download the microdata from the website

## Limits

- Too small for rare conditions or narrow subgroups
- No panel: you cannot follow a person over time
- No claims, so health care use is self-reported and coarse
- Survey weights and the clustered design must be used correctly
- Questionnaire items change across waves; pooling years takes care

Often the fastest way to establish a descriptive fact that motivates a project, before committing to an administrative data application.

SECTION 4

# Korea Health Panel Study

## Korea Health Panel Study

한국의료패널 — a household panel on health care use and spending

- Follows about **7,000 households and 21,000 individuals** over time, interviewing the same families repeatedly.
- **Sampled at the household level**, so spouses, parents, and children can be identified and studied together.
- Records health care use at the **visit level** — outpatient, inpatient, and emergency — with what each visit cost the household.
- Collects what administrative data lack: individual and household income, educational attainment, private insurance, health behaviors, and unmet need.
- Note that the panel was redesigned, so the earlier and later waves are separate panels rather than one continuous series.

# What you can study, and what you cannot

## Strengths

- **Out-of-pocket spending including non-covered care** — the gap in the claims data
- Catastrophic health expenditure and financial burden
- Unmet medical need, and why people forgo care
- Within-family spillovers and household health decisions
- Income and health over time, with the same people observed repeatedly
- Easy access: sign a consent form and the files are sent to you

## Limits

- Self-reported throughout, including diagnoses and spending
- Too small for rare outcomes, and births and deaths are few
- Health behaviors recorded in categories rather than continuous amounts
- Attrition, as in any long panel

SECTION 5

# Choosing among them

# Side by side

| | NHIS claims | K-CURE | KNHANES | KHPS |
|---|---|---|---|---|
| Type | Administrative | Registry + administrative | Survey + exam | Survey panel |
| People | 1M sample, or the full population | 2.3M cancer patients | ∼10,000 per year | ∼21,000 |
| Structure | Long panel, 2002–2023 | Panel, 2012–2022 | Repeated cross-sections | Household panel |
| Income | Premium decile only | Premium decile only | Measured | Measured |
| Education | No | No | Yes | Yes |
| Biomarkers | Screening results | Checkup results | Full examination | No |
| Diagnoses | Billing codes | Registry, with stage | Self-reported + measured | Self-reported |
| Spending | Covered care only | Covered care only | No | Including out-of-pocket |
| Families | No | No | Household roster | Yes, by design |
| Rare outcomes | Yes | Yes, for cancer | No | No |
| Access effort | High | High | **Very low** | Low |

## Which data for which question

| If your question is about··· | Start with |
|---|---|
| A rare disease, a rare adverse event, or mortality | NHIS customized |
| A policy with an age or eligibility cutoff | NHIS sample cohort |
| Cancer survival, stage, or early detection | K-CURE |
| Prevalence, trends, or undiagnosed disease | KNHANES |
| Diet, nutrition, or measured risk factors | KNHANES |
| Out-of-pocket burden or unmet need | KHPS |
| Families, spouses, or household decisions | KHPS |
| Long-term care in old age | NHIS sample cohort |

## Practical advice

- **Use the surveys to design the administrative study.** Establish the descriptive fact in KNHANES or KHPS, where data arrive in a day, before spending months on an NHIS application.
- **Check your case counts before you apply.** A rare condition in a one-million sample may not support the analysis you are proposing.
- **Write the application and the IRB protocol together**, in Korean. They must describe the same study, and English protocols have been rejected.
- **Plan your outputs in advance.** Everything leaving the analysis environment passes an export review with minimum cell size rules.
- **Call before you apply.** The portals are sparsely documented and email often goes unanswered.

# Where to read more

## Documentation I maintain

- NHIS Data Documentation
  Databases, variables, and research notes in English
- K-CURE Data Documentation
  The Cancer Public Library, table by table

## Official portals

- NHIS data sharing service
- K-CURE portal
- KNHANES
- Korea Health Panel Study

**Corrections welcome**

Both documentation sites are works in progress, written from public sources and hands-on use. If something there is wrong, please tell me.

# Thank you

**Siho Park**
**KDI School of Public Policy and Management**
**sihopark@kdischool.ac.kr**